The assignment is to update the CSCI6461 Simulator to include a number search from the terminal. A user enters 20 numbers between -32768 and 32767 followed by a target and the closest number in the array to the target is printed to the console. It utilizes the direct mapping cache so the 20 numbers are stored in cache.

Code Changes

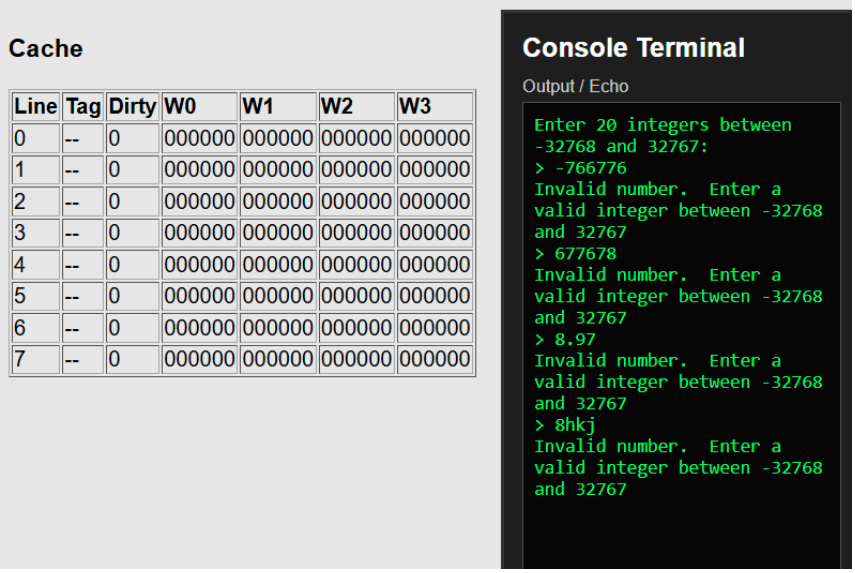
The function sendTerminalLine() was modified to read the input and write back the number search messages. After checking for valid input it calls storeWord() to store the number in memory and then increments the pointer. The 20 numbers are stored in memory locations 2000 to 2019. As a result, they are stored in the cache rows 4, 5, 6, 7, and 0. Once all 20 numbers have been entered a state variable is used to indicate the next value is the target. After a valid target is entered the 20 numbers are fetched from cache by getWord(), and the function findClosest(arr, target) is used to get the result.

New code is added in the section /*Number search code*/. The function resetNumbers() resets the pointer to the start of the memory block dedicated to storing the 20 numbers. The function inputNumber(line) checks each character of the supplied string entered at the terminal to determine if it's a valid integer between -32768 and 32767, and either returns it or NaN if it's not. The function toOctal2(n) converts decimal numbers between -32768 and 32767 to octal as described above. The findClosest(arr, target) returns the closest value to the target in the array.

The function updateCacheDisplay() was updated to display numbers between -32768 and 32767 in octal format. The first digit is 0 for positive and 1 for negative, and the remaining 5 digits are for the absolute value of the number. -32768 is 100000, -32767 is 100001, and so on. 0 is 000000, up to 32768, which is 077777. The reset button onlick code was changed to reset the number array and show the instructions for the number search.

Example

Below are screenshot illustrating the number search feature. If numbers outside the range -32768 to 32767 or decimal numbers or values with invalid characters are entered an error message appears.



Cache

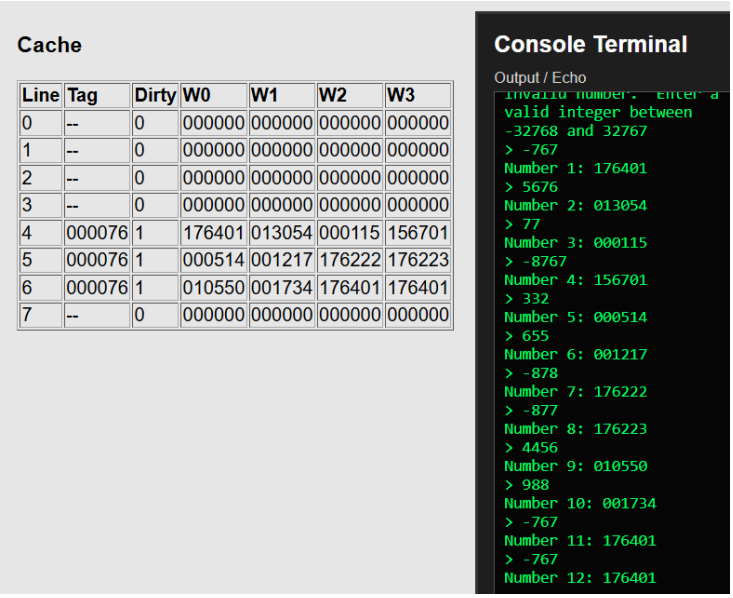
Line	Tag	Dirty	W0	W1	W2	W3
0	--	0	000000	000000	000000	000000
1	--	0	000000	000000	000000	000000
2	--	0	000000	000000	000000	000000
3	--	0	000000	000000	000000	000000
4	--	0	000000	000000	000000	000000
5	--	0	000000	000000	000000	000000
6	--	0	000000	000000	000000	000000
7	--	0	000000	000000	000000	000000

Console Terminal

Output / Echo

```
Enter 20 integers between
-32768 and 32767:
> -766776
Invalid number.  Enter a
valid integer between -32768
and 32767
> 677678
Invalid number.  Enter a
valid integer between -32768
and 32767
> 8.97
Invalid number.  Enter a
valid integer between -32768
and 32767
> 8hkj
Invalid number.  Enter a
valid integer between -32768
and 32767
```

As the 20 numbers are entered their octal representations are printed to the console and the fill cache starting from row 4,5,6,7 and then 0.



Cache

Line	Tag	Dirty	W0	W1	W2	W3
0	--	0	000000	000000	000000	000000
1	--	0	000000	000000	000000	000000
2	--	0	000000	000000	000000	000000
3	--	0	000000	000000	000000	000000
4	000076	1	176401	013054	000115	156701
5	000076	1	000514	001217	176222	176223
6	000076	1	010550	001734	176401	176401
7	--	0	000000	000000	000000	000000

Console Terminal

Output / Echo

```
Invalid number.  Enter a
valid integer between
-32768 and 32767
> -767
Number 1: 176401
> 5676
Number 2: 013054
> 77
Number 3: 000115
> -8767
Number 4: 156701
> 332
Number 5: 000514
> 655
Number 6: 001217
> -878
Number 7: 176222
> -877
Number 8: 176223
> 4456
Number 9: 010550
> 988
Number 10: 001734
> -767
Number 11: 176401
> -767
Number 12: 176401
```

Cache

Line	Tag	Dirty	W0	W1	W2	W3
0	000077	1	177677	000027	001424	000040
1	--	0	000000	000000	000000	000000
2	--	0	000000	000000	000000	000000
3	--	0	000000	000000	000000	000000
4	000076	1	176401	013054	000115	156701
5	000076	1	000514	001217	176222	176223
6	000076	1	010550	001734	176401	176401
7	000076	1	000127	023224	000003	000000

Console Terminal

Output / Echo

```
Number 10: 001734
> -767
Number 11: 176401
> -767
Number 12: 176401
> 87
Number 13: 000127
> 9876
Number 14: 023224
> 3
Number 15: 000003
> 0
Number 16: 000000
> -65
Number 17: 177677
> 23
Number 18: 000027
> 98765
Invalid number.  Enter a
valid integer between
-32768 and 32767
> 788
Number 19: 001424
> 32
Number 20: 000040

Enter target:
```

The closest number to 533 is 655

```
Enter target:
> 533
Closest number to 533 in
[-767,5676,77,-8767,332,6
55,-878,-877,4456,988,-76
7,-767,87,9876,3,0,-65,23
,788,32] is 655


Enter 20 integers between
-32768 and 32767:
```